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CEREBELLUM AND BASAL GANGLIA

----- Active space --

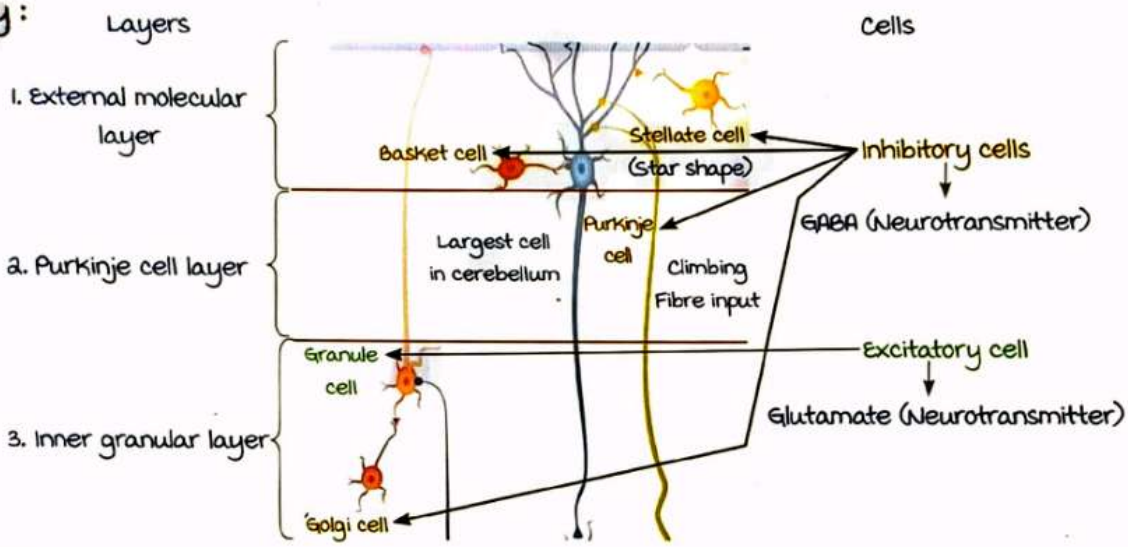
Cerebellum

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Function : Timing & coordination of motor movement.

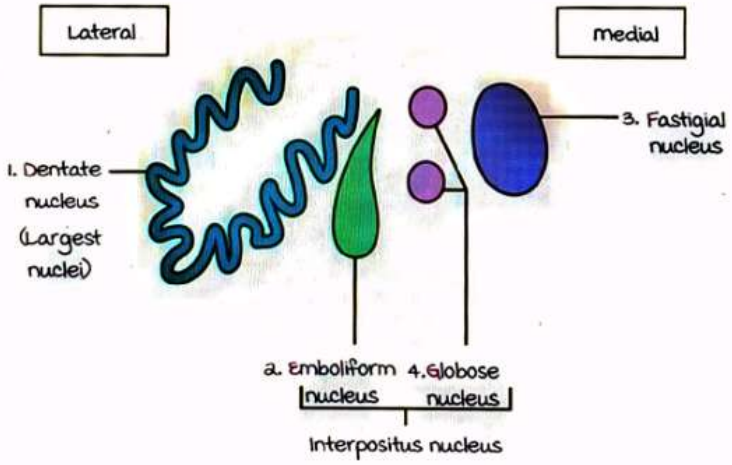
ANATOMY :

Histology :

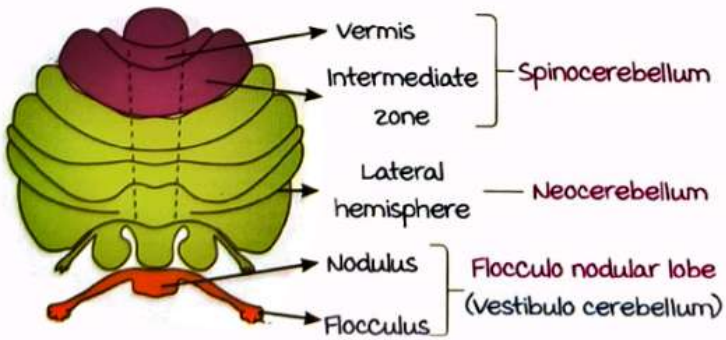


Deep nuclei

Mnemonic : D, E, F, G



Gross anatomy :



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PHYSIOLOGY :

- Sensory input ⟶ Cerebellum ⟶ Motor output.

Start-stop signal circuit :

Start pathway :

Inferior olivary nucleus ⟶ Climbing fibres (⊕) ⟶ Deep cerebellar nuclei (⊕) ⟶ Brain stem & thalamus (⊕) ⟶ Cerebral cortex ⟶ movement starts.

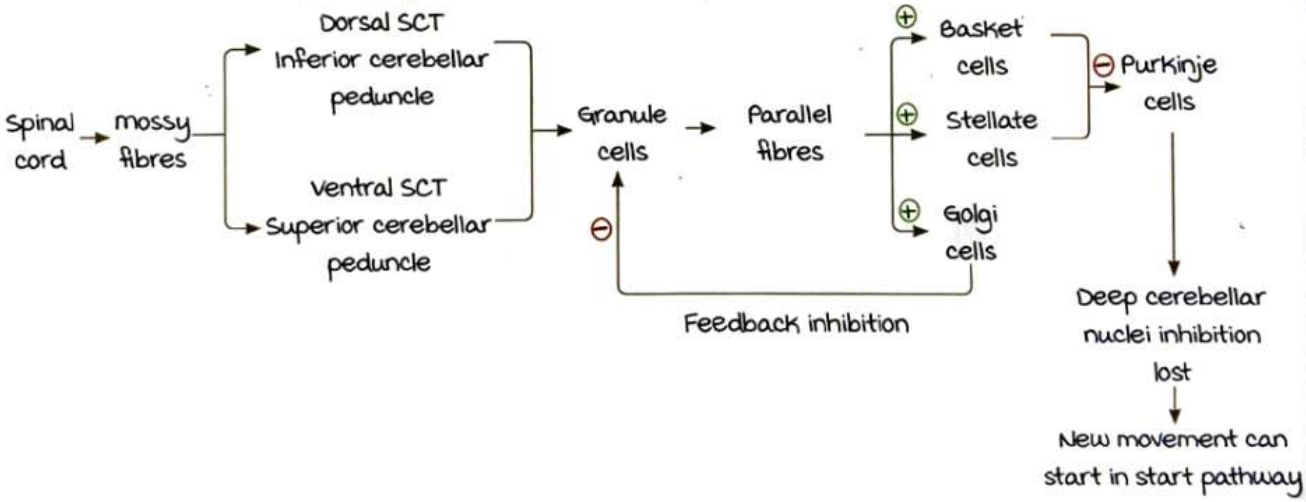
Eg : When archer pulls the string of the bow to release the arrow.

Stop pathyway :

Inferior olivary nucleus ⟶ Climbing fibres (⊕) ⟶ Purkinje cells (⊖) ⟶ Deep cerebellar nuclei ⟶ movement stops.

Eg : When the archer releases his fingers from the string (D/t inhibition of DCN).

Pre-start pathway :



Eg : For the archer to pull the string again the inhibition on the Deep cerebellar nuclei (DCN) has to be removed.

Functions :

	Nucleus	Function
Spinocerebellum	• Emboliform. • Fastigial. • Globose.	Coordination of motor movements
Vestibulocerebellum	Vestibular	• Maintaining equilibrium/balance. • Regulating eye movements.
Neocerebellum (Recently evolved)	Dentate	Planning motor movements

Feedback

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Note : Vestibular apparatus helps in maintaining balance along with cerebellum. ----- Active space -----

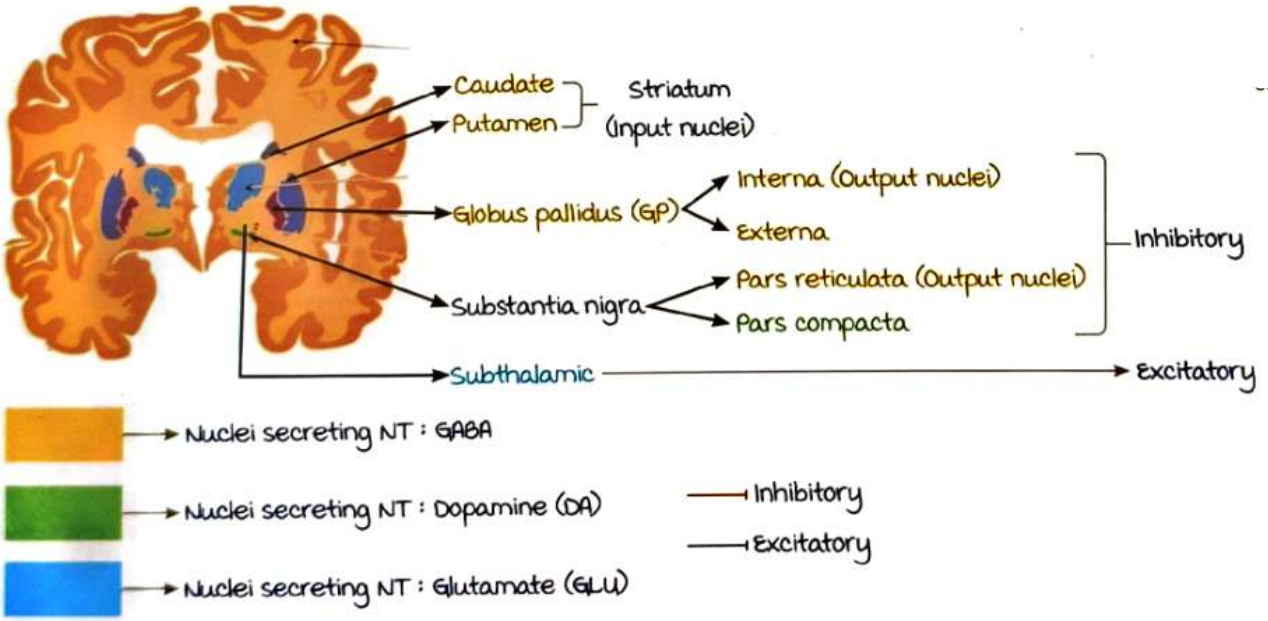
SIGNS OF CEREBELLAR LESIONS :

1. Truncal ataxia.
2. Wide based stance.
3. Titubation (Tremors of head).
4. Ataxic/drunken gait.
5. Nystagmus.
6. Decomposition of movements (Breakdown into individual parts).
7. Past pointing (Finger-nose test).
8. Kinetic/intentional tremors : Seen with activity (Absent at rest).
9. Dysarthria : Scanning speech.
10. Pendular Knee jerk.
11. Dysdiadochokinesia : Inability to perform alternating supination & pronation movements.
12. Rebound phenomenon (Unable to stop the movement).

Basal Ganglia 00:32:13

ANATOMY :

Nuclei :

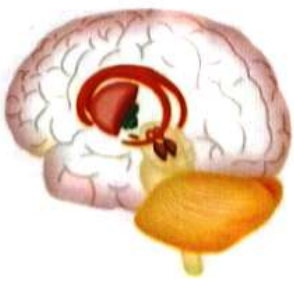


PHYSIOLOGY :

Functions :

1. Planning of motor movements.
2. Cognition (Caudate nucleus).
3. Inhibition of voluntary & involuntary movements (Even at rest).

Significance : Basal ganglia lesions → Involuntary movements.



Basal ganglia

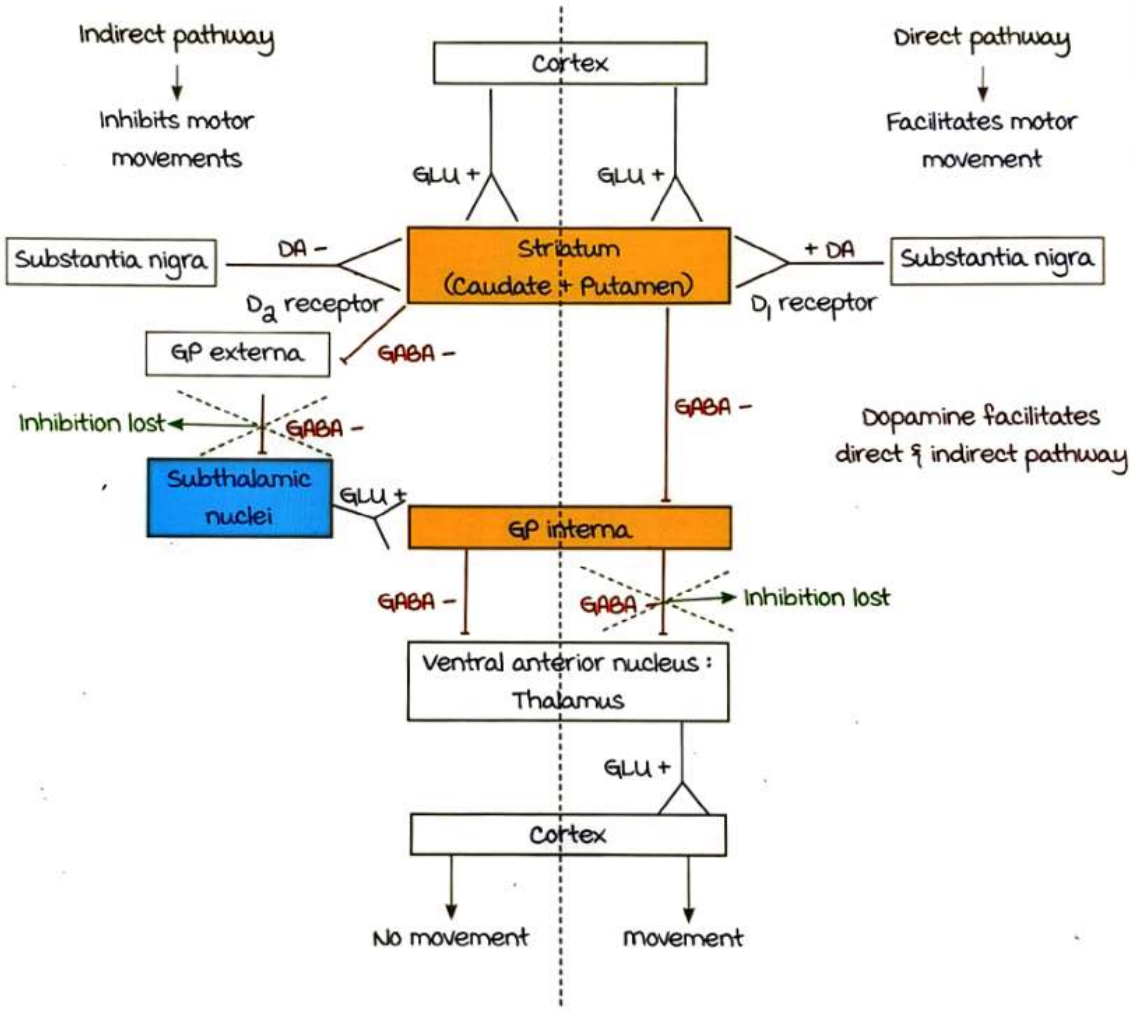
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NEURAL PATHWAY :



DISORDERS :

Parkinsonism :

Pathology : ↓ dopamine in substantia nigra (Pars compacta).

Clinical features :

1. Bradykinesia : Paucity (↓) of movements.
2. Tremors : Pill rolling tremors (Resting tremors).

Treatment :

1. Medical :

L-DOPA → Crosses BBB.

- Over time becomes ineffective (D/t progressive loss of dopaminergic neurons).

Note : Dopamine can't cross BBB.

Feedback

2. Surgical therapies :

----- Active space -----

a. Stereotactic ablation
- Thalamus :
  - Ventral anterior nucleus.
  - Ventral lateral nucleus.

↓

Alleviates symptoms

b. Deep brain stimulation
- Basal ganglia :
  - Subthalamic nucleus.

↓

Improves movements

Huntingtons disease :

Pathogenesis :

Huntingtons gene → Neuroprotective
Huntingtons gene → Anti apoptotic
Huntingtons gene → Produces brain derived neurotrophic factor

↓ mutation

↓ Trinucleotide repeat : CAG

↓ ↑ polyglutamine levels

↓ Excitotoxicity

↓ Loss of GABA releasing cell from caudate nucleus (Striatum)

↓ ↑ involuntary movements (Chorea)

Clinical features :
- Chorea : Dance like involuntary movements.
- Memory abnormality : Cognition is affected.

Involuntary movements :

Type	movement	Pathology site
Chorea	Dance like	Caudate nucleus
Athetosis	Slow writhing (Snake charmers)	Globus pallidus
Hemiballismus	Sudden violent contraction of 1 large joint (Flinging/ballistic)	Subthalamic nucleus

Feedback